	a long period of time.
7	
(a)	• Lower emission of greenhouse gases • Does not require petrol as fuel (use less non-renewable energy sources) • Quieter engines
(b)(i)	$a = v - u / t$ $= 2.72166$
	$s = v^2 - u^2 / 2a$ $= 320$ m (2 sf)
(b)(ii)1.	Units of C_d = Units of $(F_d / \rho v^2 A)$ = 1

(b)(ii)2.	Power = *Fv*
	= 1000 W (2 sf)
(c)(i)	0.28
(c)(ii)	Kinetic energy of car to electrical energy of generator to (chemical) potential energy of battery
(c)(iii)	Increase in efficiency = (0.08 * 0.80 * 0.90)
	= 0.06 (1sf)
(d)(i)	Energy stored per unit mass of the battery
(d)(ii)	Mass of battery = (30 x 1000 / 140) / 48 x 4 1.1 kg (2sf)
(d)(iii)	Time = Energy / Power = 4.3 hours (2sf)
(d)(iv)1	69.86 %
(d)(iv)2	Point plotted to half smallest square
(d)(iv)3	Best-fit line drawn
(d)(iv)4	*P* = 77.75 - 78.25 %
(e)(i)	• Electric cars consume electricity in charging which produces CO_2 at the point of power generation • Production of components of car may produce CO_2
(e)(ii)	CO_2 emission in 1 charge = 0.4 x 30 x 1000 = 12000 g
	Emission rate = 12000 / 172 = 69.76744 g km^{-1} Band A1